

**Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden
Markov Models**

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Abstract

Objective: This study proposes a method for analyzing individual learners' thought processes and investigates whether these differences can be identified. **Background:** Personalized learning environments should be tailored to how learners understand and process information. To achieve this, it is crucial to develop methods that can identify differences in thought processes. **Method:** Thirteen participants were asked to generate as many criteria as possible for classifying six geometric figures into two groups. Participants' thinking styles were assessed through a questionnaire. A method combining eye-tracking data collected during the task and a hidden Markov model (HMM) was proposed to analyze the thought processes. **Results:** Participants exhibited distinct response patterns, which could be grouped into two categories: one group classified the figures based solely on their visual attributes, while the other employed more imaginative criteria that extended beyond the visual attributes. Differences in response patterns were associated with hierarchical thinking style. Although there was no significant difference in gaze fixation durations based on scores for hierarchical thinking style, differences were observed in the temporal patterns of gaze transitions, which could be identified with 80% accuracy using HMMs. **Conclusion:** Learners' thought processes differences are reflected in the temporal dynamics of fixation scan path transitions and can be identified using HMMs. **Application:** This study demonstrates the potential for indirect analysis of cognitive processes using eye-tracking, offering a non-invasive alternative to direct brain measurements. The proposed method may serve as a foundational technology for future systems that analyze learners' cognitive strategies and comprehension in educational contexts.

Keywords: eye tracking, hidden Markov model, individual differences, problem-solving

Précis: This study proposes a method to analyze learners' thought processes using eye-tracking and hidden Markov model. Experimental results show differences in thought processes

52 can be identified with high accuracy, suggesting potential applications for personalized learning
53 systems.

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Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden

Markov Models

Introduction

In today's rapidly evolving society—exemplified by innovations such as Generative AI—children are increasingly required to adapt flexibly to change (Antoninis et al., 2020). Accordingly, the need to cultivate a wide range of skills and competencies has become more critical than ever. Achieving this goal calls for educational practices that are tailored to the characteristics of each learner (Dosch & Zidon, 2014; Moallemi, 2024; Turner et al., 2017). However, there is currently no consensus on which learner characteristics should be prioritized, highlighting the need for deeper investigation into individual learning processes.

Within the field of learning sciences, which explores how people learn, it is widely argued that instruction should be adapted to the unique ways in which each learner constructs understanding. Understanding, in this context, is defined as the process of integrating new information into pre-existing knowledge structures and subsequently restructuring them (Davis, 1984; Sawyer, 2014a, 2014b). However, since each learner's prior experiences and existing knowledge differ, their methods of achieving understanding also vary, highlighting the importance of personalized instructional approaches (Coffield et al., 2004; Miyake, 1986; Peach et al., 2021; Shirouzu et al., 2010). Yet, quickly identifying a learner's preferred approach to comprehension remains a significant challenge, especially in contexts where teachers and students have not yet developed rapport. In the absence of such understanding, learners may struggle to fully grasp new concepts, potentially limiting the effectiveness of instruction. Therefore, it is essential to create personalized learning environments that incorporate systems capable of identifying how each learner understands.

Establishing such environments requires robust models for analyzing the diverse cognitive approaches that learners use to make sense of information. Specifically, this involves investigating the thought processes unique to each learner. In this study, we define cognition as “the broad mental process of acquiring and understanding knowledge through perception, memory, judgment, and decision-making,” and thinking as “the conscious process of manipulating mental representations, such as ideas and images, to perform reasoning and problem-solving” (American Psychological Association, n.d.). We focus specifically on thought processes. However, because these processes are internal and not directly observable, distinguishing among them remains a major challenge. This necessitates the development of analytical models capable of inferring and differentiating thought processes in a quantifiable manner.

One widely used method for analyzing cognition is the Think Aloud protocol, in which participants verbalize their thoughts while performing a task (Ericsson & Simon, 1993). Although this method can provide valuable insights, it is limited in its ability to capture non-verbalized or unconscious cognitive activity (Cowan, 2019; Ericsson & Fox, 1993). Other methods such as functional MRI (fMRI), near-infrared spectroscopy (NIRS), and electroencephalography (EEG) offer non-invasive measurements of brain activity, but their susceptibility to motion artifacts and operational complexity make them less suitable for use in typical educational settings (Janssen et al., 2021; Maclaren et al., 2014; Seghier et al., 2019; Zhao et al., 2023).

In recent years, eye tracking has emerged as a promising alternative for investigating cognitive processes (Eraslan et al., 2015; Lin et al., 2017; Nakano & Ishii, 2010; Salvucci & Anderson, 1999; Wang et al., 2018; Zawoyski et al., 2015). Grounded in the eye–mind hypothesis—which posits that gaze patterns reflect underlying cognitive states (Just & Carpenter, 1980; Lai et al., 2013; Rodrigues & Rosa, 2019; Schindler & Lilienthal, 2019; Wu & Liu, 2022)—eye tracking has been used to assess cognitive load (Kanaan, 2021; Shafiei, 2024), evaluate student performance (Kim, 2024),

and inform the design of educational environments (Eisma, 2020; Kim, 2023). Notably, recent studies have also applied eye tracking to analyze cognitive processes more directly. For example, Hsing et al. (2022, 2023) demonstrated that specific gaze patterns correspond to distinct spatial problem-solving strategies, and Bittner et al. (2024) successfully classified learning styles from eye-tracking data using machine learning with 69% accuracy. Similarly, our prior study (Ohata et al. (2024)) have demonstrated the effectiveness of eye-tracking as a method for analyzing cognitive processes.

These findings support the effectiveness of eye tracking to reveal learners' thought processes. On the other hand, to analyze learners' thought processes, analyzing system is required that can visualize thought processes and distinguish their differences quantitatively. While prior research has highlighted specific differences in cognitive processes, the development of computational analyzing models capable of identifying and distinguishing thought processes remains limited. Moreover, previous studies have not sufficiently examined whether it is feasible to analyze thought processes with the direct aim of identifying individual learners' ways of understanding. Therefore, the objectives of this study are (1) propose a method for analyzing the differences and characteristics in learners' thought processes, and (2) evaluate whether these differences can be detected using the proposed method.

Methods

Participants

Thirteen healthy adult volunteers (both male and female) participated in this study on a voluntary basis. Two participants were excluded from the analysis due to failure to follow task instructions, resulting in a final sample of eleven participants (mean age $\pm$ SD = 22.7 $\pm$ 6.17 years). A previous eye movement study reported a significant trend toward comparable individual differences

in problem-solving strategies, using a sample size of similar magnitude to that of the present study (Shannon et al., 2008). All experimental procedures were approved by the Ethics Review Committee of the University of Tokyo (Approval No. 22-416) and conducted in accordance with the approved protocols. Written informed consent was obtained from all participants prior to participation.

Task

The task was to classify six different shapes (shown in Table 1) into two groups and finding as many classification criteria as possible. The shapes systematically varied across four visual dimensions: color (red or blue), pattern (solid or striped), size (40 × 40 px or 200 × 200 px), and geometric form (circle, square, or triangle). Each shape was presented individually on a tablet interface (see Figure 1b). Participants were instructed to manipulate the shapes directly on the touchscreen to form groupings according to their self-defined criteria. The task continued until the participant could no longer generate new classification categories.

This task design was inspired by prior research in personality theory, specifically the "color–form problem" developed by Honkavaara (1958a, 1958b), which has been used to assess perceptual and cognitive style. The color–form problem examines how individuals judge the similarity of visual stimuli, focusing on whether they prioritize color or geometric form in their decisions. Previous findings suggest that individuals who prioritize color may exhibit higher levels of creative thinking (Honkavaara, 1958a, 1958b). Building upon this theoretical framework, the present study introduces a novel variant of the task to investigate individual differences in thought processes.

Table 1*Parameters of Figures*

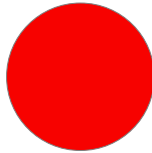
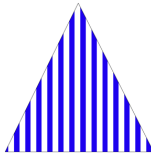
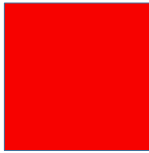
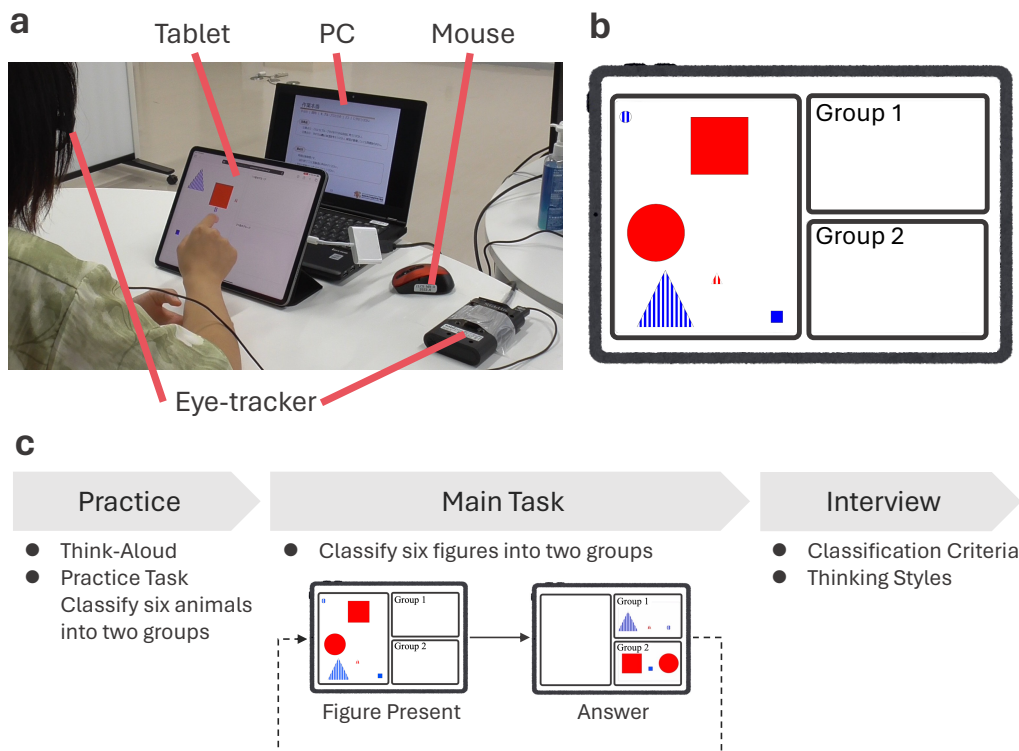
Index		1	2	3	4	5	6
Appearance							
Parameters	Shape	Circle	Triangle	Square	Circle	Triangle	Square
	Color	Blue	Red	Blue	Red	Blue	Red
	Pattern	Stripe	Stripe	Solid	Solid	Stripe	Solid
	Size	Height: 1cm, Width: 1cm			Height: 5cm, Width: 5cm		

Figure 1*Experiment. a Experimental Setup. b An Example of Tablet Screen. c Experiment Procedure*

Investigation of Cognitive and Thinking Characteristics

Participants were also asked to complete the Thinking styles (Coffield et al., 2004; Sternberg, 1988; Sternberg & Zhang, 1999) questionnaire in order to investigate individual differences in thought processes. Thinking styles refer to the preferred ways individuals organize, direct, and manage their thinking. This concept was proposed in Sternberg's theory of mental self-government (Sternberg, 1988), which suggests that individual thought processes are governed by the self much like a government governs a society. Using this analogy, Sternberg defined Thinking styles as five distinct categories and thirteen specific types. Unlike other psychometrics that assign a single label to a participant's cognitive or thinking profile, this approach provides a multidimensional characterization by combining several indicators of strengths and weaknesses. Moreover, a study conducted with university students in Hong Kong and the United States confirmed that cultural differences in Thinking styles are minimal (Sternberg & Zhang, 1999).

The five styles and thirteen types of Thinking styles are shown in Table 2. Each item is rated on a seven-point scale, with higher scores indicating a stronger inclination toward the corresponding style. Thinking styles does not categorize participants into one of thirteen distinct styles. Instead, each participant receives a score for all thirteen styles, allowing an analysis of the balance between their strengths and weaknesses across these dimensions. In this study, we used the simplified Japanese version of the Thinking Styles Inventory (Hiruma, 2003).

Table 2*Thirteen Items of Thinking Styles*

Dimension	Style	Description
Functions	Legislative	Like to create their own rules
	Execute	Like to follow rules
	Judicial	Like to evaluate rules and procedures
Forms	Monarchic	To be motivated by a single goal or need at a time
	Hierarchic	To be motivated by a hierarchy of goals, with the recognition that not all goals can be fulfilled equally well and that some goals are more important than others
	Oligarchic	To be motivated by multiple, often competing goals of equal perceived importance
	Anarchic	To be motivated by a potpourri of need
Level	Global	Prefer to deal with relatively large and abstract issues
	Local	Often like concrete problems requiring detail work
Scope	Internal	Tend to be introverted, task-oriented, aloof, socially less sensitive, and interpersonally less aware than externalists
	External	Tend to be extroverted, people oriented, outgoing, socially more sensitive, interpersonally more aware than internalist
Leaning	Conservative	Like to adhere to existing rules and procedures
	Progressive	Like to go beyond existing rules and procedures

Experimental Setups

The experimental setup is shown in Figure 1a. The task was conducted using a tablet (Apple iPad Pro 12.9 inch) placed in front of the participants. The eye movements of each participant during the task were recorded using an eye tracker (Tobii Pro Glasses 3). A PC was used to explain the experiment and answer a questionnaire about the experiment. Participants were positioned face-to-face with an experimenter.

Procedure

The experimental protocol comprised three sequential stages: a practice session, a main task session, and a post-task interview, as depicted in Figure 1c. During the practice session, participants engaged in a think-aloud familiarization and completed a practice task designed to resemble the main task. Initially, they were explained the think-aloud method, including its purpose

and procedures, followed by a brief practice to familiarize themselves with the approach. In the practice task, participants classified six animal icons displayed on the tablet into two categories in as many distinct ways as possible. Eye-tracking data were collected during this stage to ensure proper calibration and participant comfort with the apparatus. In the main task, participants were asked to classify six geometric shapes into two groups, generating as many classification categories as possible. The shapes were displayed individually on the tablet. Participants were explicitly informed that there were no constraints on classification criteria and no time limit. Participants continued to think aloud during the task, and their eye movements were recorded. Following task completion, participants engaged in a semi-structured interview in which they were asked to describe the rationale behind each classification they had made. After the interview, participants completed a post-experiment questionnaire as well as the Thinking Styles Inventory.

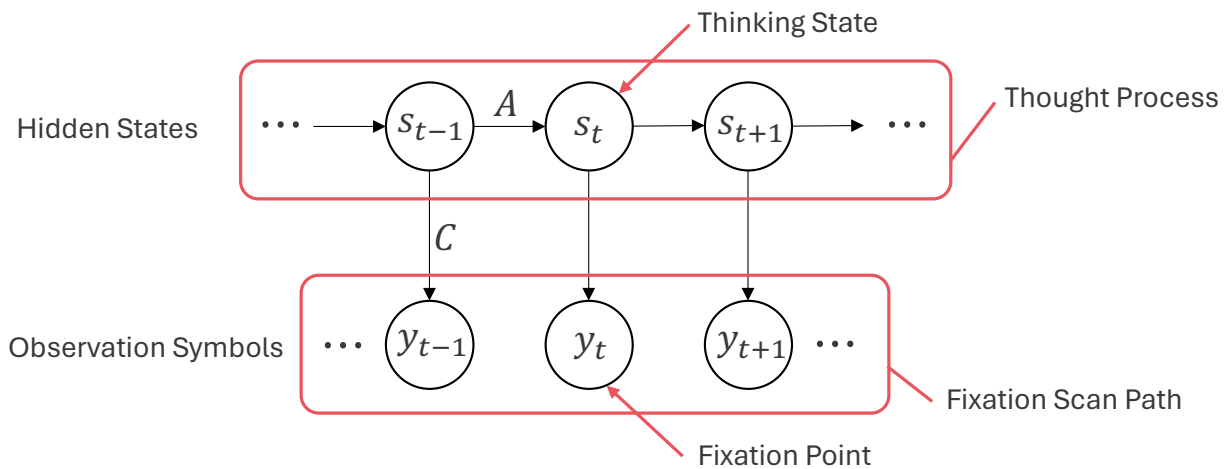
Data Collection and Statical Analysis

Four primary data types were collected in this study: task responses, thinking style scores, fixation duration data, and scan path data. The think-aloud data were used as a supplementary reference to inform the analysis. Task responses consisted of the multiple classification criteria generated by each participant in response to the figure classification task. Thinking styles were assessed using participants' scores across thirteen dimensions. To explore the relationship between thinking styles and thought processes, participants were grouped according to patterns observed in their classification responses. For each group, mean scores across the thirteen thinking styles were computed. To identify statistically significant differences between groups, two-tailed independent samples t-tests were conducted for each thinking style. The significance threshold was set at $\alpha = 0.05$. Fixation and scan path data were collected using the eye-tracking system. A fixation was defined as a continuous gaze maintained on a specific Area of Interest (AOI) for 300 milliseconds or longer, in accordance with previous research (Solso, 2015). To account for individual variability in

overall fixation behavior, fixation durations were z-transformed within participants prior to statistical analysis. To investigate the relationship between gaze behavior and thinking styles, z-transformed fixation duration data were analyzed using a linear mixed-effects model (LMM). In this model, figure type and thinking style group (based on problem-solving strategy classification) were included as fixed effects, along with their interaction term. Participant was included as a random intercept to control for repeated measures and inter-individual differences. Scan path data represented the sequential transitions of gaze between AOIs, capturing the temporal dynamics of visual exploration. These data were modeled using a Hidden Markov Model.

Figure 2

Graphical Representation of a Hidden Markov Model



Analyzing Thought Process Using Scan Path and Hidden Markov Models

To analyze learners' thought processes, we employed a Hidden Markov Model (HMM) in combination with scan path data. HMMs are statistical models commonly used to analyze time-series data. The graphical representation of an HMM is illustrated in Figure 2. In HMM, the subsequent state s_{t+1} is determined solely by the current state s_t . This property is known as the

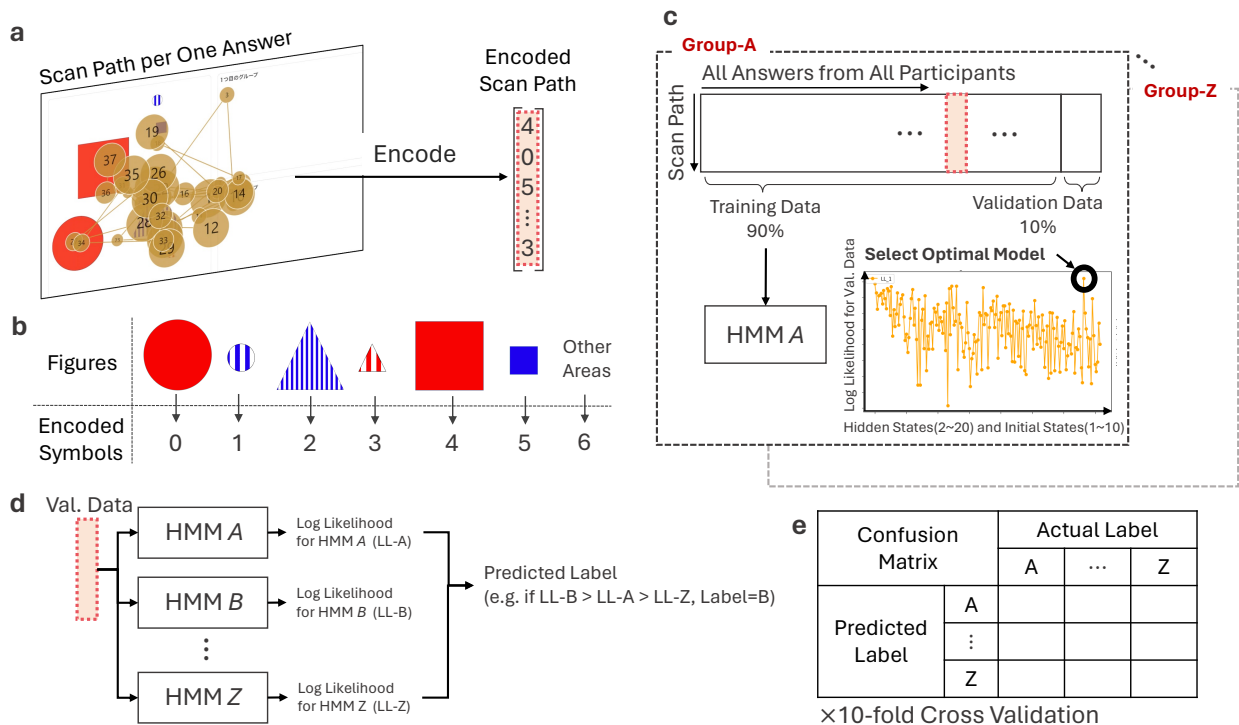
Markov process implying that earlier states do not directly influence future transitions. Because the states in an HMM are unobservable, the model estimates these hidden states based on observed symbols y_t . During the experiment, participants need to observe six figures while considering possible classification criteria. According to the eye-mind hypothesis (C. J. Wu & Liu, 2022; Just & Carpenter, 1980; Lai et al., 2013; Rodrigues & Rosa, 2019; Schindler & Lilienthal, 2019), eye movement patterns reflect cognitive processes. Therefore, by modeling the fixation scan path using an HMM, we aim to infer and analyze the underlying thought processes of participants.

In this study, the sequence of observation symbols $\mathbf{y}_{1:T} = y_1, y_2, \dots, y_T$ represents the fixation scan path, where each y_t corresponds to a fixation point on a specific Area of Interest (AOI). The hidden states $\mathbf{s}_{1:T} = s_1, s_2, \dots, s_T$ represent the underlying thought processes, with s_t denoting the latent thinking state at time t . The model further includes three parameters of probability distributions: the state transition probabilities A , which define the likelihood of transitioning from one hidden state to another; the symbol emission probabilities C , which represent the likelihood of observing a particular fixation given a hidden state; and the initial state probabilities $\boldsymbol{\pi}$, which define the probability distribution over the starting states. The hidden states s_t transition with probabilities specified in the rows of A , and at each time step emits an observation symbol y_t according to the probabilities in the rows of C .

There are several reasons why an HMM is particularly well-suited for analyzing thought processes. For one, because the focus of this study is on the process of thinking, it is necessary to analyze time series data. With an HMM, the time series data of both the thought process and the fixation scan path can be modeled as probability distributions of state transition A and symbol output C probabilities. Moreover, while other models such as RNNs or LSTMs exist, they typically require large amounts of data and are more computationally intensive. In contrast, the HMM provides a lighter analysis.

Figure 3

Analyzing Method for Identification of Differences in Thought Process. **a** Extracting Scan Path Per One Answer. **b** Encoding Rule for Fixation Sequences. **c** Training the Hidden Markov model (HMM). **d** Model Inference from Validation data. **e** Evaluation and Cross Validation.



Construction of the Fixation Scan Path Dataset

Figure 3a, 3b illustrate the method used to construct the dataset. We constructed the dataset for the HMM by extracting the data $y_{1:T}$ from fixation scan-paths recorded during tasks. Figure 3a represents the fixation scan path from when a problem is presented until the participant begins moving the figures while considering classification criteria. The brown circles indicate fixation points. The size of each circle indicates the duration of the fixation; larger circles mean a longer fixation. We created the dataset by applying the encoding rule shown in Figure 3b, extracting the fixation point sequence from the overlapping areas of the brown circle and the figure. Each figure is

assigned a unique numerical label, and regions outside the displayed figures are also assigned numbers.

Model Fitting and Validation Procedure

An HMM can be constructed by estimating the model parameters $\theta = (\pi, A, C)$. The commonly used method is the Baum-Welch algorithm, but this algorithm often leads to local optima and overfitting (Yang et al., 2003; Beal, 2003). Variational Bayesian Expectation Maximization (VB-EM) addresses this by using priors to reduce redundant states and estimate model complexity automatically (Beal & Ghahramani, 2003). In this study, we employed VB-EM as a training algorithm. This study treated the observation symbol as categorical variables and set the model parameters using a categorical distribution. The prior distribution was set as a Dirichlet distribution, which is the conjugate prior of the categorical distribution. Additionally, the prior values were set as uniform for A and C , while π was initialized randomly.

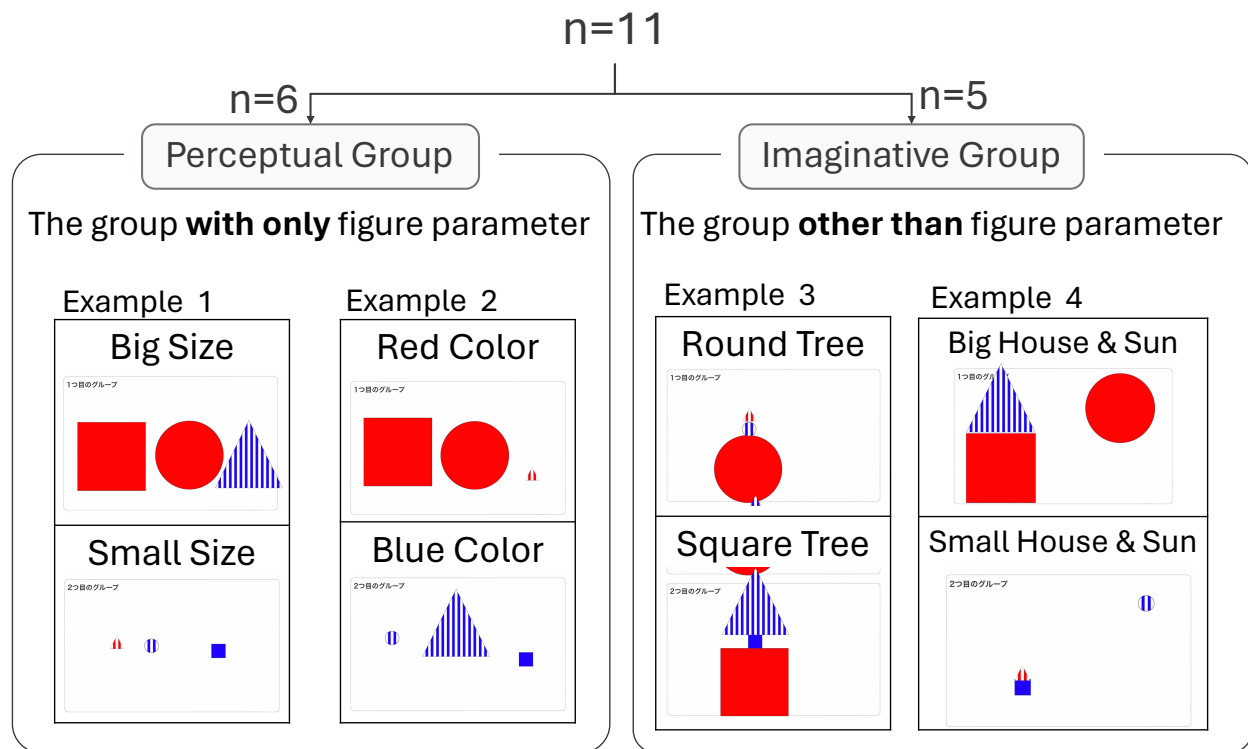
The method for fitting the model using fixation scan paths is illustrated in Figure 3. Fixation scan paths datasets were created using all answers all participants (Figure 3a, b). Next, participants were grouped based on differences in thinking styles associated with problem-solving approaches, and an HMM along with a corresponding dataset was constructed for each group. Each dataset was divided into training data and validation data, and the each HMM was trained on the training data (Figure 3c). Specifically, models were trained with combinations of the number of hidden states (2 to 20) and the number of initial states (1 to 10), and the log-likelihood of the validation data for the trained models was calculated. The optimal model was selected as the one with the parameters that maximize the log-likelihood of the validation data (Figure 3c). Next, the validation data from all groups are input to all the trained models to output predicted labels (Figure 3d). The predicted label was determined as the one for which the log-likelihood of the validation data was highest among the models (Figure 3d). This allowed us to infer which model most likely generated the validation data.

The accuracy, recall, and F1 score were calculated for the output predicted labels (Figure 3e). All the above was conducted using 10-fold cross-validation (Figure 3e).

Results

Participant Response Classification

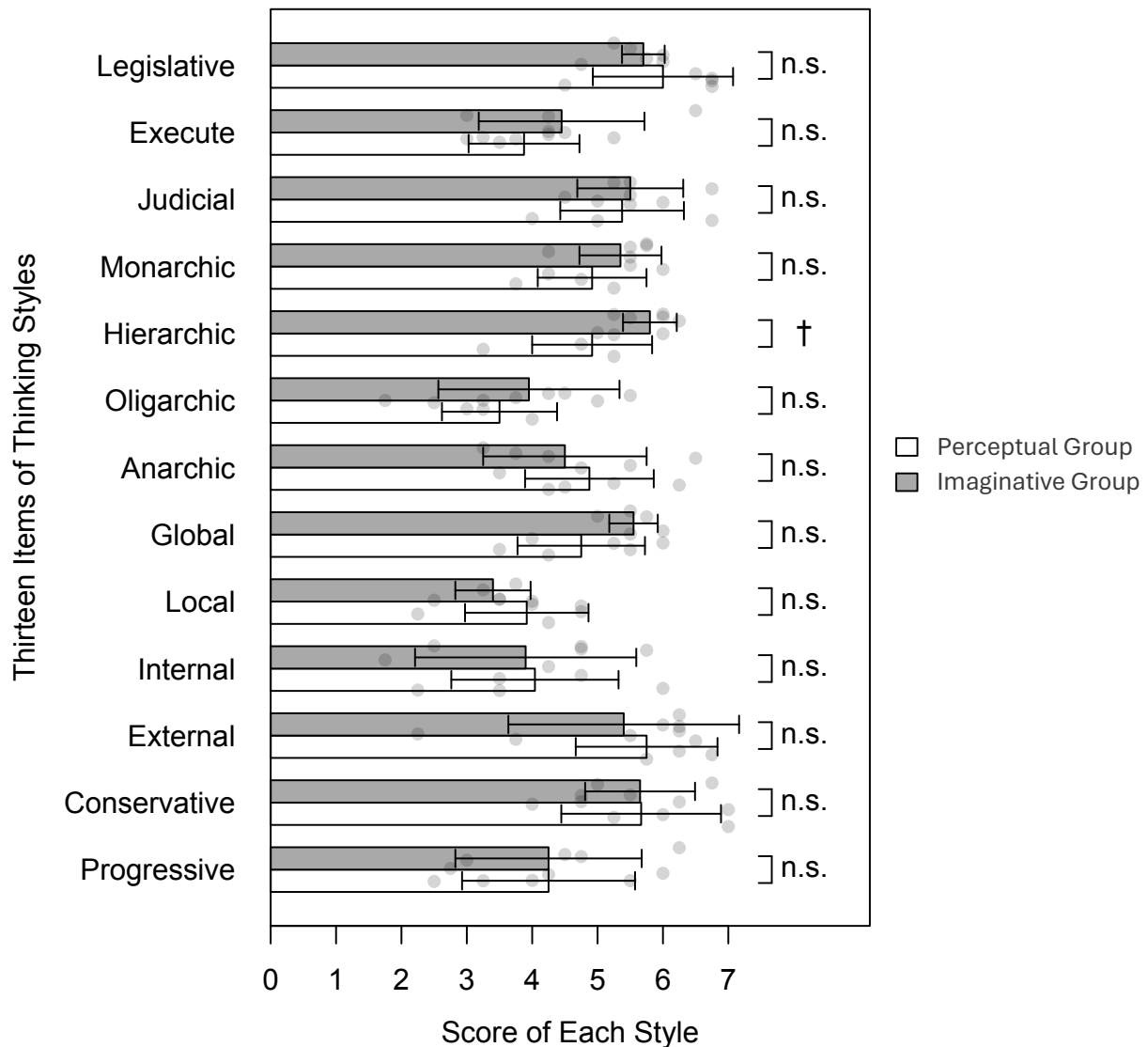
A total of 97 responses were collected from 11 participants. Based on the tendencies of their classification criteria, participants were categorized into two groups, as illustrated in Figure 4. The first group ($n = 6$) provided classification strategies grounded in the physical properties of the stimuli specifically, color, pattern, size, and geometric features such as corners or lines, as outlined in Table 1. These strategies included responses based on single parameters or combinations thereof. This group is hereafter referred to as the Perceptual Group. The second group ($n = 5$) provided answers that were not only related to the figure parameters, but also to the relatively free imagination. These responses included classifications based on combinations or relationships among figures, associations with symbolic or character-like features, visual impressions, and the spatial relationship between objects. This group is referred to as the Imaginative Group.

Figure 4*Differences in Answers to the Task***Thinking Style Comparison**

To explore the relationship between individual differences in thinking style and problem-solving approaches, mean scores for 13 thinking styles were compared between the perceptual and imaginative group. As shown in Figure 5, a trend toward significance was observed for the hierarchical thinking style, with the perceptual group exhibiting lower scores than the imaginative group ($t(7) = 2.12, p = .07, d = 0.39$). No significant differences emerged for the remaining 12 styles. Hierarchical thinking style, as defined in Sternberg's theory (1988), reflects an individual's tendency to organize goals in a structured hierarchy, prioritizing them based on importance. These results demonstrate that perceptual group participants and imaginative group may differ in their thinking approaches, particularly in terms of hierarchical thinking.

Figure 5

Comparison of Average Scores across Groups for the Thirteen Items of Thinking Styles



Fixation Duration Analysis

To examine the effect of thinking style associated with problem-solving approach on fixation duration, we constructed a linear mixed-effects model with thinking style group (High-Hierarchical vs. Low-Hierarchical), figure type (Index #1-6 as shown in Table1), and their interaction as fixed effects. Participant was included as a random intercept to account for inter-individual

variability. Participants were divided based on their hierarchical thinking scores: those scoring above the group mean were assigned to the High-Hierarchical group ($n = 6$), and those below the mean to the Low-Hierarchical group ($n = 5$). Model results indicated no significant main effect of hierarchical thinking style on fixation duration ($\beta = -0.17, SE = 0.33, t(48) = -0.52, p = .609$). In contrast, a significant main effect of figure type was observed: participants exhibited longer fixations for the figure #5 (blue, triangle, stripe, big) ($\beta = 1.81, SE = 0.33, t(48) = 5.43, p < .001$) and the figure #6 (red, square, solid, big) ($\beta = 1.51, SE = 0.33, t(48) = 4.56, p < .001$) compared to the reference figure#1 (blue, circle, stripe, small). Other figure types did not significantly differ from the reference. No interaction between thinking style and figure type reached significance ($ps > .05$). These results suggest that fixation duration is modulated primarily by visual characteristics of the stimuli, rather than by differences in hierarchical thinking style.

Table 4

Fixed Effects from Linear Mixed-Effects Model Predicting Fixation Duration by Thinking Style and Figure Type

Fixed Effect	Estimate (SE)	t (df)	p
Intercept	- 0.623 (0.235)	- 2.65 (48)	.011*
(Low-Hierarchical x figure1(blue, circle, stripe, small))			
Group (High- Hierarchical Thinking Style)	- 0.171 (0.332)	- 0.52 (48)	.609
Figure Type			
2 (red, triangle, stripe, small)	- 0.100 (0.332)	- 0.30 (48)	.764
3 (blue, square, solid, small)	- 0.021 (0.332)	0.06 (48)	.951
4 (red, circle, solid, big)	0.540 (0.332)	1.62 (48)	.111
5 (blue, triangle, stripe, big)	1.806 (0.332)	5.43 (48)	<.001***
6 (red, square, solid, big)	1.514 (0.332)	4.56 (48)	<.001***
Group x Figure Type			
2 (red, triangle, stripe, small)	- 0.014 (0.470)	- 0.03 (48)	.976
3 (blue, square, solid, small)	0.354 (0.470)	0.75 (48)	.455
4 (red, circle, solid, big)	0.724 (0.470)	1.54 (48)	.130
5 (blue, triangle, stripe, big)	0.334 (0.470)	0.71 (48)	.481
6 (red, square, solid, big)	- 0.369 (0.470)	- 0.79 (48)	.436

Scan Path Analysis Using Hidden Markov Models

To further investigate the effect of thinking style associated with problem-solving approach on scan paths, we analyzed participants' scan paths using Hidden Markov Models (HMMs). Following the procedure illustrated in Figure 3, separate HMMs were trained for the High-Hierarchical and Low-Hierarchical groups using 10-fold cross-validation. The classification performance yielded an average accuracy of 0.80 ($SD = 0.12$), average recall of 0.82 ($SD = 0.14$), and average F1 score of 0.84 ($SD = 0.10$), indicating that HMMs successfully distinguished between group-specific scan path dynamics.

Table 5

Ten-Fold Cross Validation of Fixation Scan Path Based on Differences in Thinking Styles

Metrics	<i>M</i>	<i>SD</i>
Accuracy Rate	0.80	0.12
Recall	0.82	0.14
F1-Score	0.84	0.10

Next, full dataset HMMs were trained for each group, and both state transition matrices and symbol emission distributions were extracted (see Figures 6a and 6b). The number of hidden states was fixed at seven to match the number of AOIs. To evaluate the structural centrality of hidden states, we generated weighted directed graphs from the transition matrices and applied the PageRank algorithm (Page et al., 1999). As shown in Table 6, in HMM on Low-Hierarchical group, State 0 and State 1 serve as network centers, whereas in HMM on High-Hierarchical group, State 3 and State 5 are comparatively more central. These structural differences are also evident in the graph structures displayed in Figure 7. These findings indicate that the temporal dynamics of scan paths are reflected in differences among the HMM hidden states and identified between groups.

Table 6*PageRank Centrality of Hidden States in Each Group Scan Paths*

States	PageRank	
	Low-Hierarchical	High-Hierarchical
0	0.330	0.249
1	0.334	0.391
2	0.065	0.026
3	0.029	0.119
4	0.086	0.039
5	0.029	0.091
6	0.124	0.083

Hidden Markov Model Transition Matrices and Symbol Output Distributions for the Low-Hierarchical and High-Hierarchical Thinking Style Groups

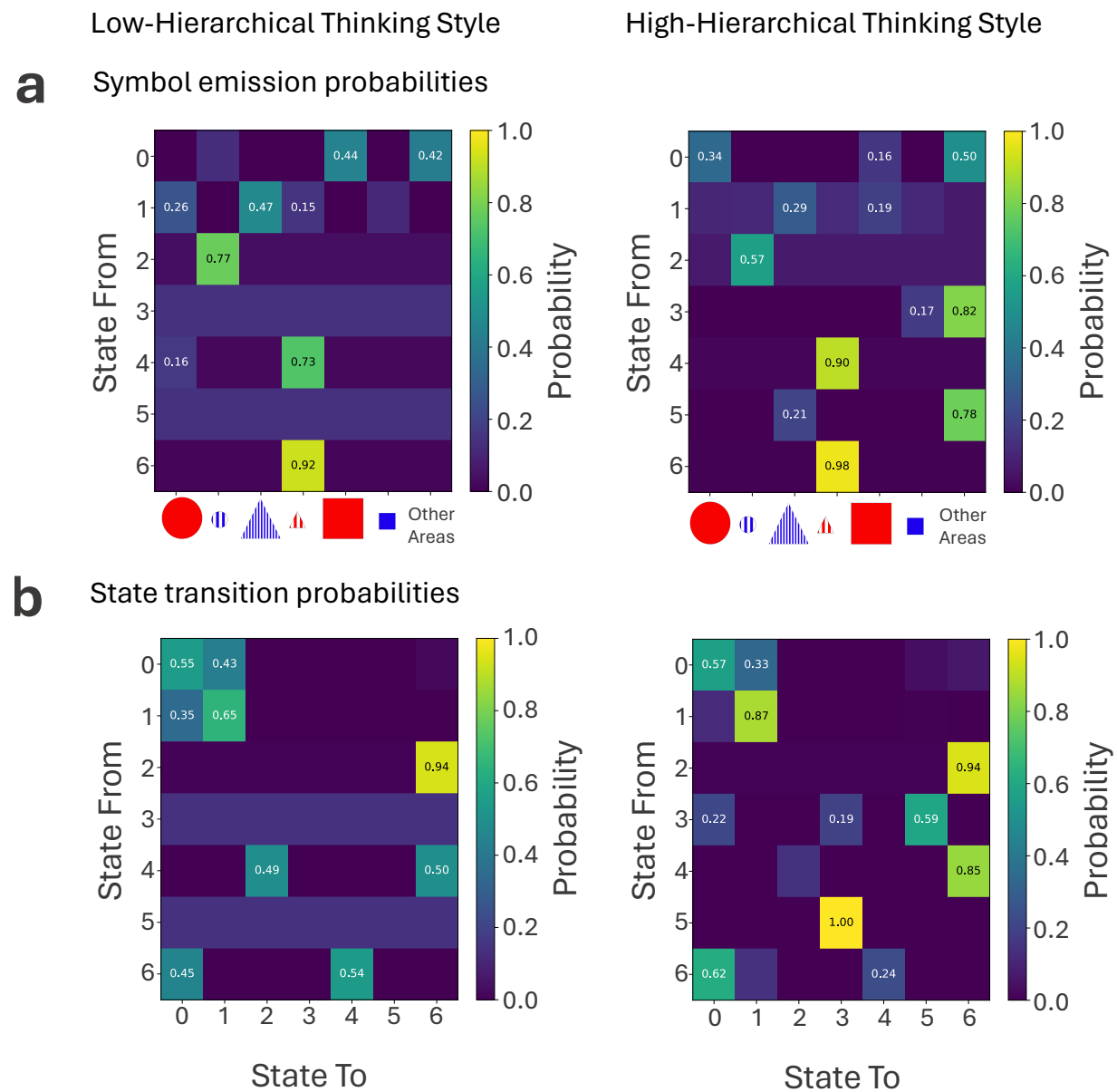
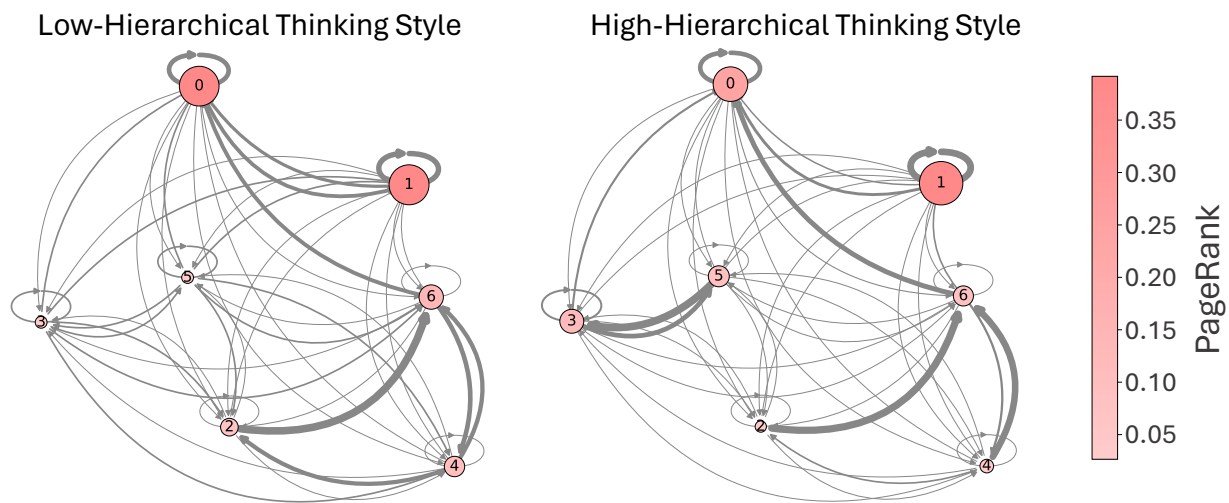


Figure 7

Graph Representations of State Transition Matrices for Low-Hierarchical and High-Hierarchical Thinking Style Groups. Node Size Reflects PageRank Centrality, and Edge Thickness Corresponds to Transition Probability



Discussion

In this study, participants repeatedly classified six different figures into two groups. As a result, their responses were categorized into two patterns: a Perceptual Group (based on the visual attributes of the figures) and an Imaginative Group (based on more creative interpretations).

Participants in the imaginative group exhibited a stronger tendency toward a hierarchical thinking style. Although there were no significant differences in fixation durations on individual figures depending on thinking style, differences in scan path patterns were observed between the groups, and these patterns could be identified with 80% accuracy using Hidden Markov Models (HMMs).

These differences in response tendencies and thinking styles may reflect the operation of distinct thought processes among learners. According to Sternberg, "A hierarchical style refers to the tendency to solve problems by organizing them in a gradual and structured manner." Participants

with a stronger hierarchical thinking style were more likely to provide creative responses. For example, they might have interpreted a triangle as the "roof of a house," suggesting that they associated figure parameters with their own knowledge and mental imagery. These findings indicate that, even when engaging in the same task, individuals may employ different thought processes depending on their thinking styles.

Although no significant group differences in fixation duration were found, fixation duration varied depending on the visual characteristics of the figures. These results suggest that bottom-up attentional mechanisms particularly those driven by visual saliency (e.g., figure size, color, and pattern) strongly influenced gaze behavior. Prior research has shown that stimulus attributes such as size and surface complexity modulate fixation time (Nuthmann et al., 2016), and similar effects appear to underlie the present findings. Therefore, in this study, figure size and the presence or absence of patterns may have influenced fixation durations, suggesting that the differences might be due to visual saliency rather than differences in thought processes.

By contrast, the scan path analysis using Hidden Markov Models (HMMs) revealed clear distinctions between groups. The models successfully classified participants' gaze trajectories with high accuracy (80% on average), indicating that differences in thought processes were embedded in the temporal dynamics of eye movements. This finding aligns with prior work demonstrating that scan path structures vary with task type and problem-solving strategy (Guo et al., 2024; Laurence et al., 2018). These findings suggest that scan path patterns are influenced by top-down attention driven by task demands, and that such patterns differ across groups. Thus, even with the same task, learners may engage in different thought processes. These individual differences are not captured by static indicators like fixation duration but are instead reflected in dynamic indicators such as scan paths, which can be visualized and classified using time-series statistical models such as HMMs.

This study has several limitations. First, although differences were observed not only in answers but also in cognitive and thinking characteristics through the experiment and analysis, these findings may differ depending on the age group and sample size of the participants. Therefore, at this time, it is necessary to continuously verify the trends seen in the answers and cognitive and thinking characteristics of new participants. To address this, future research should focus on developing generalized models that can predict how differences in cognitive and thinking characteristics are reflected in variance in problem-solving answers. Second, the proposed method for analyzing thought processes showed that thought processes are identifiable and the differences can be expressed by the probability distribution of HMMs, but it remains unclear what specific features characterize human thought processes. In response to this, it is conceivable that by integrating data on eye movements and brain activity, the correspondence between gaze patterns and neural substrates can be clarified and thought processes can be modeled.

The feature of this proposal is that it demonstrates that it is possible to "indirectly" analyze thought processes using a glasses-type eye tracking device, rather than using methods such as EEG, NIRS, and fMRI that "directly" measure brain function. Analysis of thought processes in traditional learning environments has relied on externalized expressions such as notes and verbalization (Ridgway & Cox, 2024). In contrast, in recent learning environments, it is essential for teachers to understand learners' thinking patterns early on. Therefore, systems that analyze how learners understand content are expected to become increasingly important in the future. In addition, minimizing the constraints imposed on learners is expected to become one of the important system requirements in school education. The proposed method is promising as a basic technology for systems that analyze understanding methods because it reduces physical constraints compared to brain function measurement technology and is thought to not hinder learners' thinking. The introduction of such a system is expected to have the following effects: learners will be able to

understand their own thinking tendencies and select a learning method that suits them, and educators will be able to teach in a way that is easier for learners to understand.

Conclusion

The purpose of this study was to propose a method for analyzing the differences and characteristics of each learner's thought process and to evaluate whether these differences can be identified. Although all participants were asked to generate as many different criteria as possible for dividing six geometric figures into two categories, their responses varied, suggesting that different thinking styles may be associated with distinct thought processes. While fixation duration did not significantly differ between groups defined by differences in thinking styles associated with problem-solving approaches, gaze scan paths differed depending on those groups and could be classified with 80% accuracy using Hidden Markov Models. This opens up bright prospects for the realization of a system that analyzes learners' unique thinking tendencies and ways of understanding.

Key Points

- Participants performed geometric figures classification task while their eye movements were recorded.
- An analyzing method for identifying different thought processes utilizing eye movement scan paths modeled with hidden Markov models was proposed.
- Differences in learners' thought processes were embedded in their eye movement patterns, enabling group classification with 80%
- The proposed method may serve as a foundational technology for personalized learning systems that evaluate the way each learner comprehends.

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